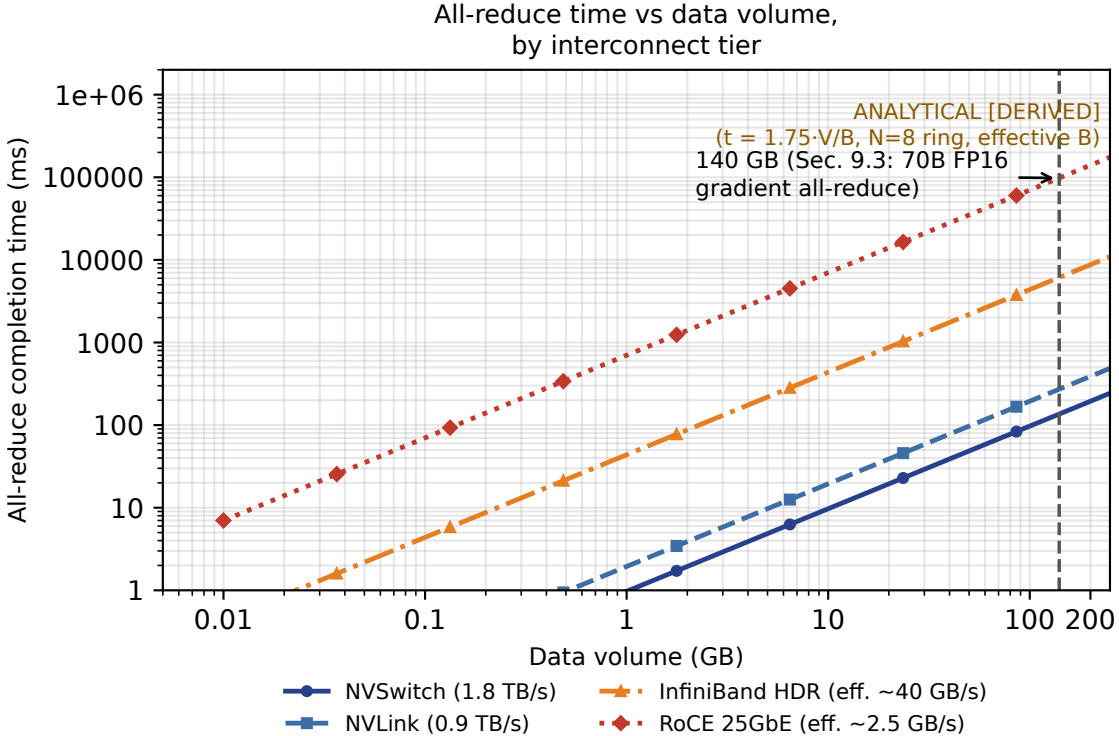
All-reduce time vs data volume,
by interconnect tier
All-reduce completion time (ms)
1e+06
100000
10000
1000
100
10
1
0.01
0.1
1
10
100
200
Data volume (GB)
ANALYTICAL [DERIVED]
(t = 1.75·V/B, N=8 ring, effective B)
140 GB (Sec. 9.3: 70B FP16
gradient all-reduce)
NVSwitch (1.8 TB/s)
NVLink (0.9 TB/s)
InfiniBand HDR (eff. ~40 GB/s)
RoCE 25GbE (eff. ~2.5 GB/s)